# TESSERA :
# An integrated HSM to bring Silicon-Level Key Security to the Server Tray

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# Agenda

- The problem with centralized HSM architectures
- Key management antipatterns across the industry
- iHSM and the TESSARA specification
- Reference architecture: attestation, provisioning, and trust model
- Standards alignment: Caliptra, LOCK, DC-SCM, SPDM
- Digital sovereignty and regulatory impact
- How to get involved

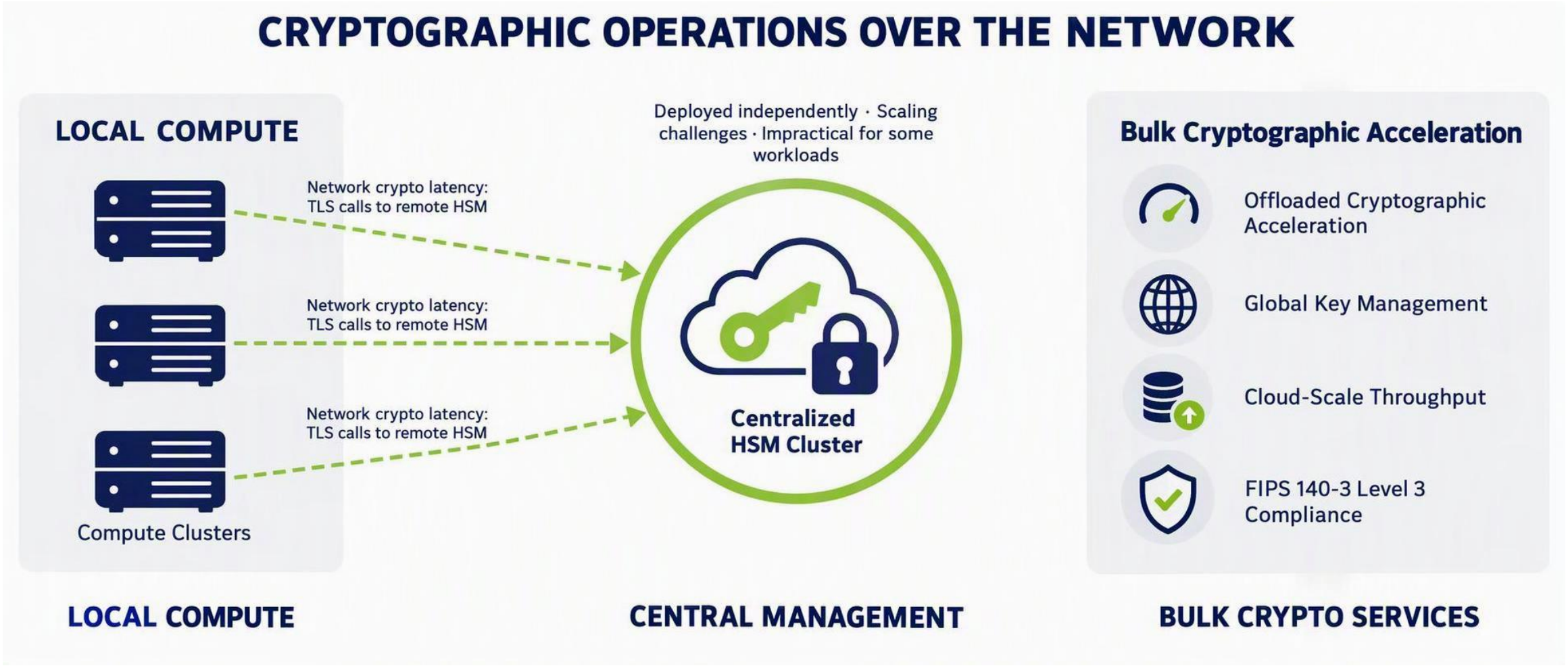
# The challenges of Centralized HSMs

- AI and confidential-compute workloads are scaling cryptographic demand faster than centralized HSM clusters can serve it
- Every operation is a network round trip to a remote vault
  - Latency, scaling bottlenecks, and an expanded attack surface
- Under pressure, teams adopt workarounds to hit performance targets
- The workarounds — not the HSM itself — are where trust quietly erodes
  - CVM enclave susceptible to SCA and SW bug
  - Certs and Keys still live in plaintext in enclave instead of dedicate HSM boundary

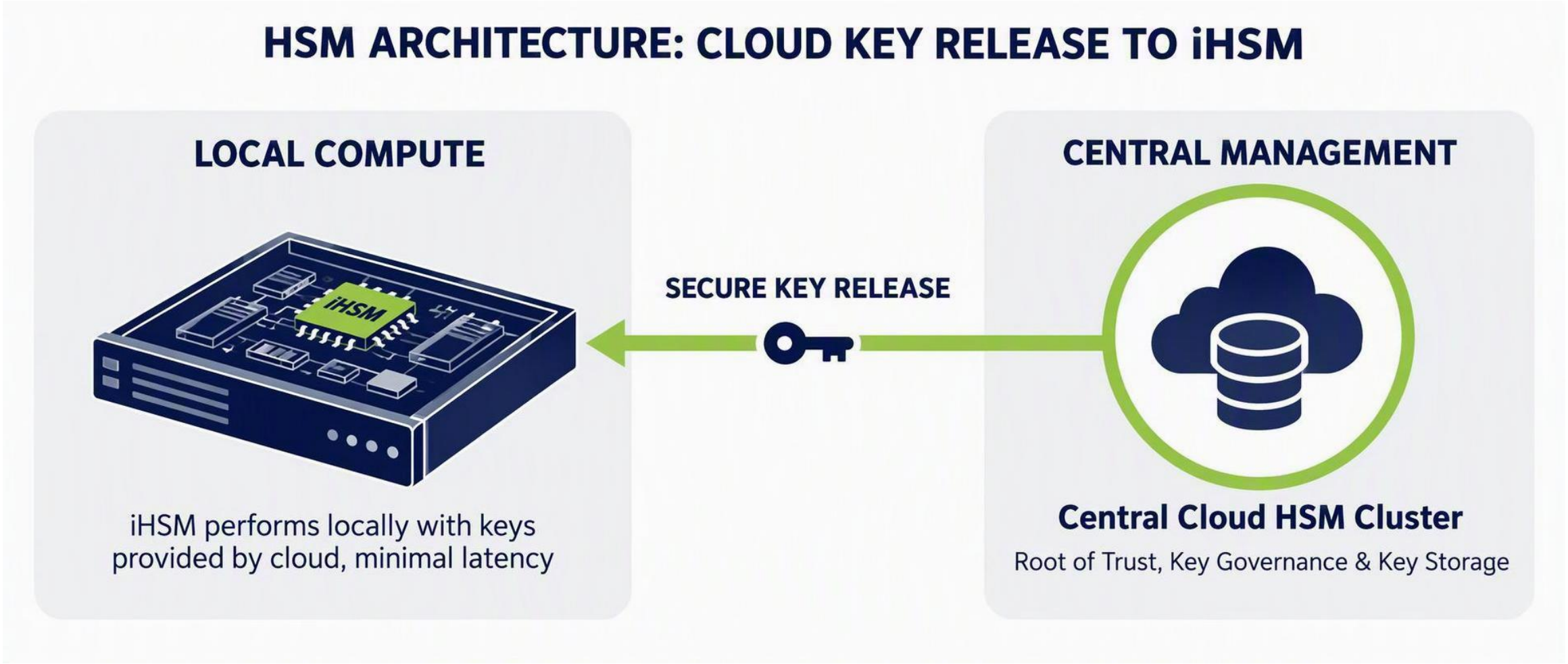
# Key Management Antipatterns

- Critical keys generated outside the HSM boundary
- Key export into workload environments with weaker assurance
- Local key caching for resilience, bypassing the HSM
- Latency-driven workarounds that weaken isolation
- Governance sprawl across over-shared security domains
- Keys can't move across vendors or clouds without re-generation, proprietary transport, or plaintext exposure risk

# Traditional HSM Architecture



# HSM Architecture: Secure Key Release to iHSM



# TESSARA: An Open iHSM Standard

*An open OCP standard that defines the iHSM*

**Unveiled today at OCP Global Summit 2026 — jointly developed by Marvell, Microsoft, and Oracle**

### FIPS 140-3 Level 3

A dedicated security module co-located with the workload in the server tray

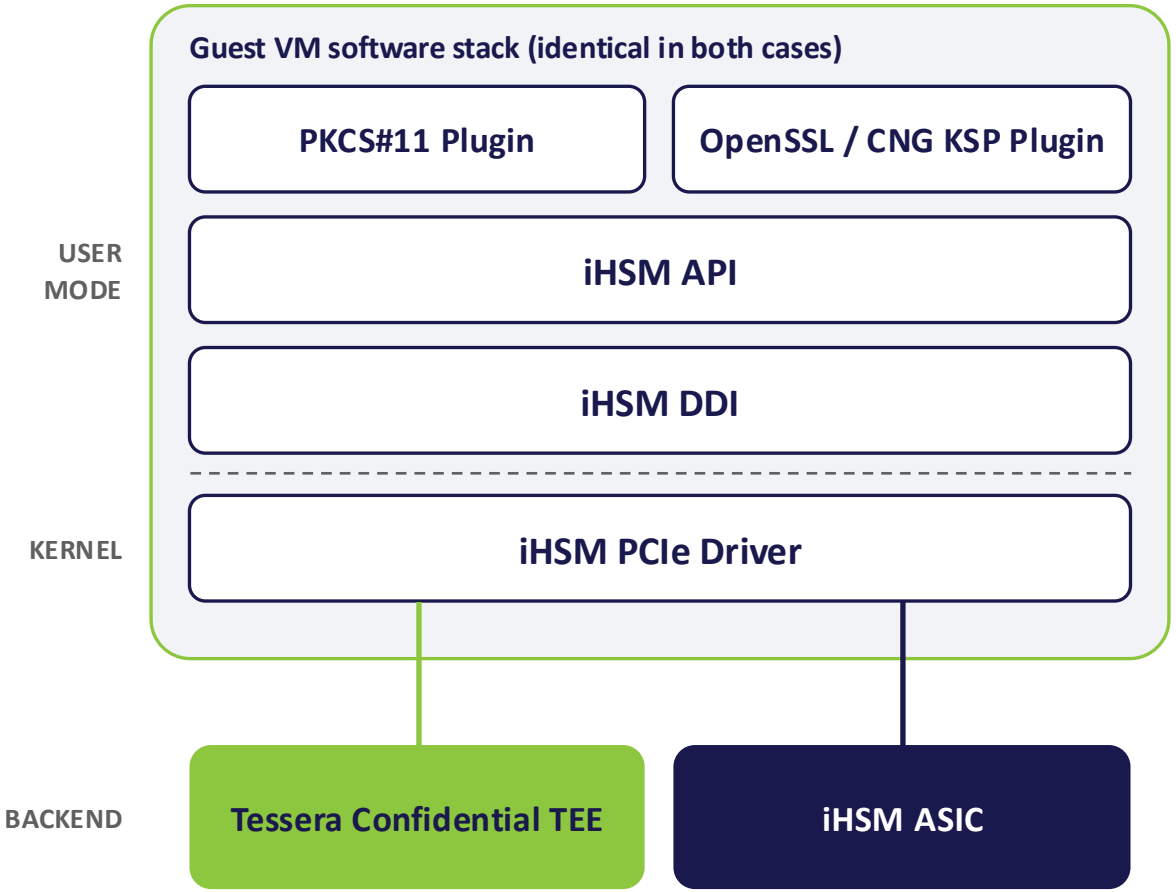
### Open-Silicon Model

The same open model Caliptra brought to hardware root of trust, now for keys

### Extends Central HSM

Extends the centralized HSM trust boundary so keys are maintained and used locally on the tray, next to the workload

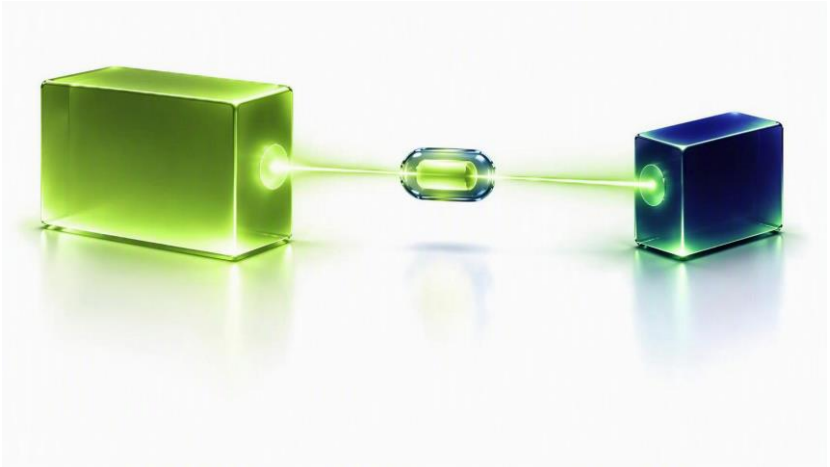
# Tessera Stack




- **PKCS#11 Plugin:** Standard PKCS#11 interface so existing applications use the iHSM unchanged
- **OpenSSL / CNG KSP Plugin:** Connects the iHSM to OpenSSL and the Windows CNG Key Storage Provider
- **iHSM API:** Common user-mode library for key management and cryptographic operations
- **iHSM DDI:** Device driver interface: the defined command boundary into the kernel
- **iHSM PCIe Driver:** Kernel driver that carries DDI commands to the backend over PCIe
- **Tessera Confidential TEE:** iHSM in firmware inside an isolated TEE, with no ASIC required
- **iHSM ASIC:** Dedicated hardware iHSM on the server tray

# DARC

## Standard PQ Safe Protocol to move keys between HSMs.



SOURCE

### Central HSM

Owns the key. Mints a single-use transfer key and wraps under it.

**ATTESTED HPKE SEALED Key**

TARGET

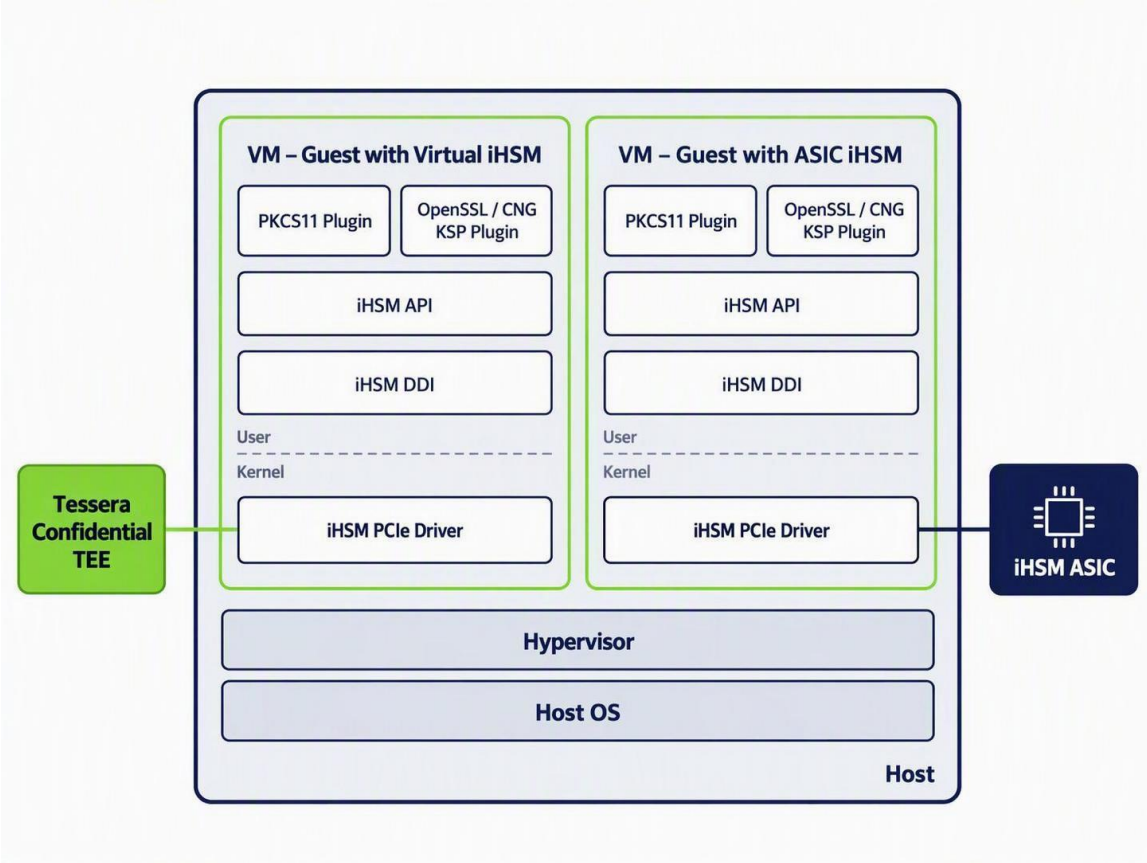
### Integrated HSM

Unwraps into its own masked-key domain, attributes preserved.

Different make, different model, no shared manufacturer chain — and the key is never marked extractable.

# TESSARA In Confidential TEE

- iHSM stack can run in an isolated TEE
- Enables iHSM functionality in firmware
- Identical guest software stack
- Allows:
  - Early enablement and deployment of the iHSM stack independent of the ASIC
  - Validation in CI/CD pipelines
  - Harmonized environments with and without iHSM ASIC



# Standards Alignment

**Caliptra**
Open-source hardware root of trust

**OCP LOCK**
NVMe key-management block

**DC-SCM**
Secure management controller & platform root of trust

**SPDM / TDISP**
Attestation and secure device interfaces

**FIPS 140-3 Level 3**
NIST cryptographic module compliance

**Post-Quantum Ready**
Hybrid (ML-KEM/ECDHE) and (ML-DSA/ECDSA)

# Digital Sovereignty and Compliance

### Data Residency
Localized trust boundaries meet data residency and jurisdiction requirements

### Regulatory Compliance
Supports DORA, OSFI, PCI-HSM, FedRAMP, and eIDAS

### Auditable Lineage
Cryptographic lineage stays intact across multi-cloud and DR environments

### Exit-Ready Controls
Key attributes, usage policy, and provenance preserved across vendor boundaries

# Founding Collaborators




LiquidSecurity® HSM silicon and firmware




Azure Integrated HSM and cloud-scale deployment




Oracle Cloud Infrastructure Key Management

*OCP Security Project working group — open invitation to hyperscalers, OEMs, silicon vendors, standards bodies, and regulated enterprises*

# Call to Action: Join Tessera

- Problem to solve
  - Keys can't move across independently operated HSM environments without re-generation, proprietary transport, or plaintext-exposure risk
- How to get involved
  - Join the TESSARA workstream through the OCP Security Project — working group kickoff follows OCP Global Summit 2026
- Where to find additional information
  - OCP Security Project: opencompute.org/projects/security
  - Azure Integrated HSM overview: aka.ms/project-tessera-ocp
  - Open-silicon model: caliptra.io

# Thank You!

**Thank you — let's build the open standard for cloud hardware security together.**

OCP Security Project: opencompute.org/projects/security

Azure Integrated HSM: aka.ms/AzureIntegratedHSM

Open-silicon model: caliptra.io

